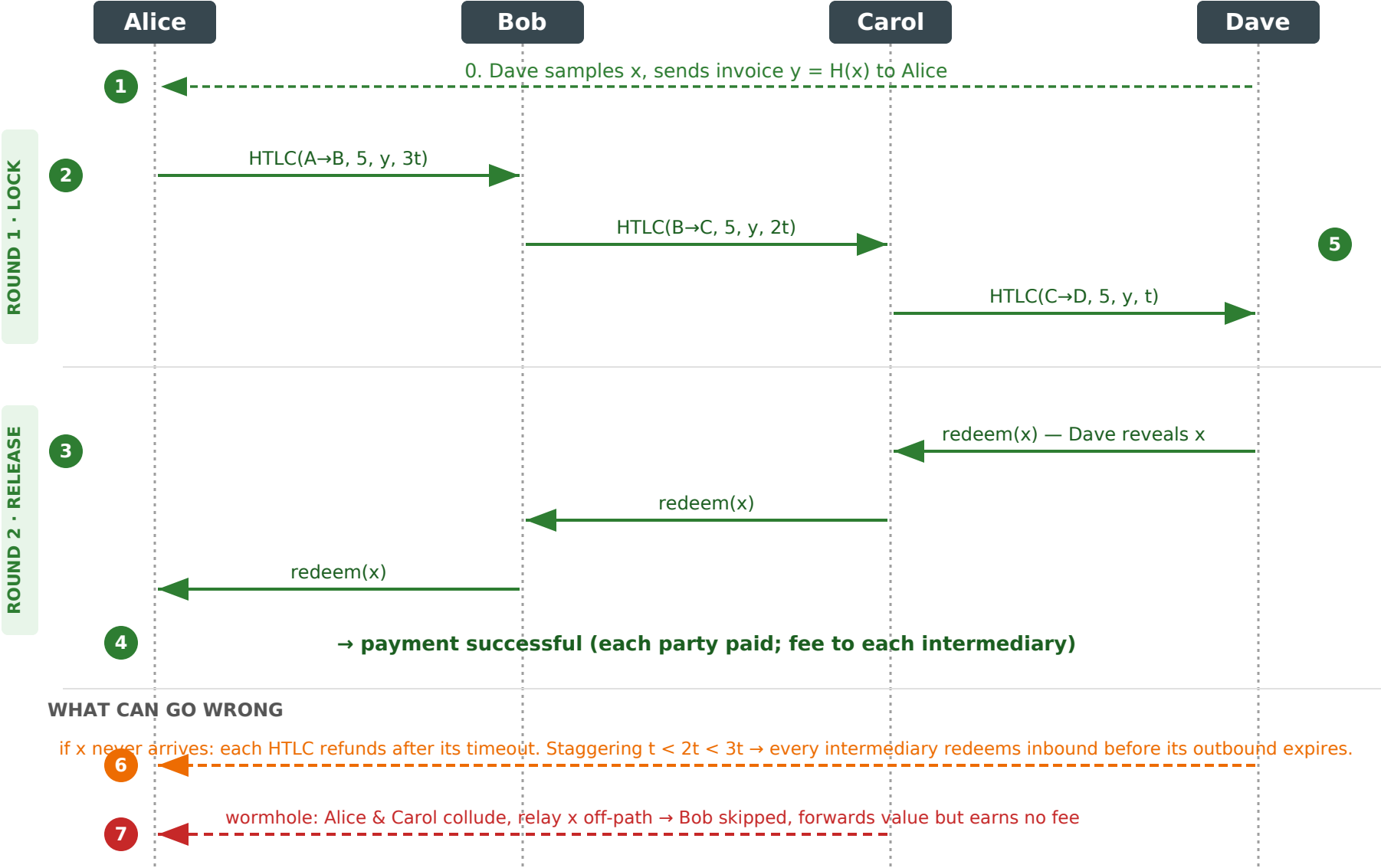


Multi-hop payment: Alice → Bob → Carol → Dave

annotated with multihop.spthy lemmas — see rail →



8 STAGGERED TIMELOCKS (block height →)



Rough draft — multihop.spthy lemmas mapped to the 4-party flow. Next: cltv_blocks / Clock / timeout.

LEMMAS · multihop.spthy

1 Invoice & preimage authentication

Invoice_Authenticates_Settlement · Preimage_Secret_Until_Rel
Forged_Invoice_Requires_Key_Compromise · Invoice_Released

2 Lock causality (forward needs previous)

Forward1_Requires_Offer · Forward2_Requires_Forward1
HTLC_On_Opened_Channel ✓

3 Release causality (redeem needs preimage)

Fulfill_Requires_Forward2 · Claim_Requires_Release
Settle_Requires_Receiver_Release ✓

4 Atomicity + non-vacuity

Payment_Atomicity_Under_Liveness ✓
Multihop_Payment_Possible ✓ (witness)

5 Value & fee conservation

EndToEnd_Value_Conservation · Fee_Conservation_Hop1/2
Fee_Strictly_Positive_Hop1/2 · Receiver_Paid_Invoice_Amount

6 Refund / griefing

Refund_Requires_Timeout · Loss_Requires_Inaction ✓
Honest_Intermediary_Refunds ✓ (reachable refund/DoS)

7 Wormhole fee theft

Wormhole_Fee_Theft_Reachable ✓
Wormhole_Steals_Exactly_The_Fees ✓ (theft = path fees)

8 Intermediary safety (under liveness T2b)

Intermediary_Never_Loses_Under_Liveness ✓
Timeout_Race_Blocked ✓ (race blocked with T2b)

✓ = machine-checked in multihop.spthy (43/43).
green = all-traces safety · orange = benign/reachable
red = attack shown reachable (by design, no PTLC).

Channel-layer lemmas (open/update/close/punish)
live on the Alice↔Bob figure.